DysphagiaSense

AI-Based Swallow Risk Screening tool

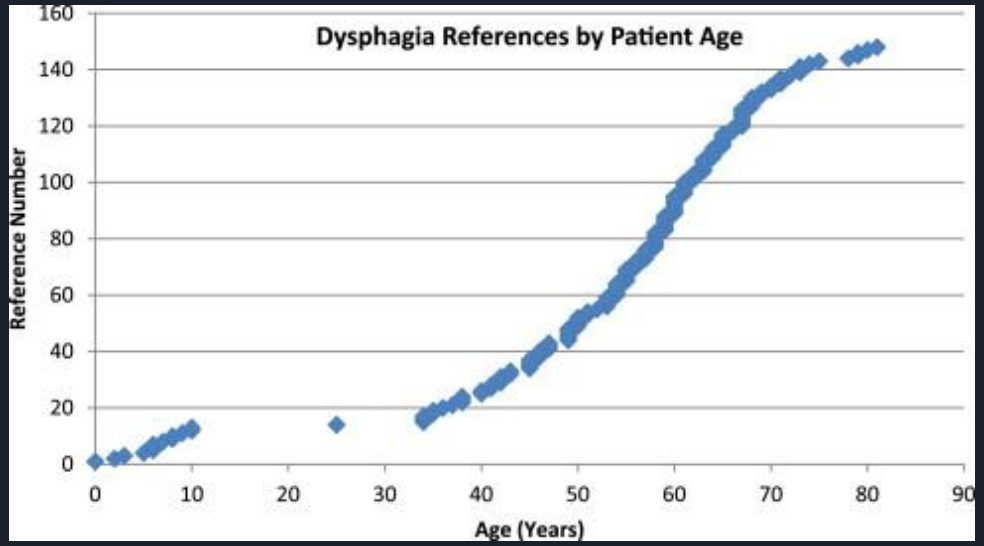
What is Dysphagia?

A swallowing disorder that makes it difficult to safely move food or liquid from the mouth to the stomach.

Who is most affected?

- Older adults are significantly more likely to develop dysphagia
- Affects up to 15% of the elderly population
- Risk increases due to:
  - Aging muscles
  - Neurological conditions (stroke, Parkinson’s)

The Hidden Risk of Dysphagia



- Over **50%** of stroke patients experience it
- Early signs are **subtle** and **often missed**
- Cases rise sharply after **age 50**
- Highest prevalence in **elderly populations**
- Aging is a major contributing factor
- Can lead to **choking** and **aspiration pneumonia** (lung infection)

The Gap in Current Screening

- Require **clinical expertise**
- Need **specialized equipment**
- Existing devices can cost **around $1000+**
- Often require sensors placed **under the chin (inconvenient and uncomfortable)**
- Not accessible for **everyday use**

Introducing **DysphagiaSense**

AI-Based Audio Risk Screening

- Records a short **swallow audio sample**
- Extracts key **acoustic features**:
  - Duration of swallow
  - Number of sound events (peaks)
  - Post-swallow activity
  - Signal patterns (energy + frequency)

Rule-Based Risk Scoring System

- Assigns **weighted scores** based on detected patterns
- Identifies indicators such as:
  - Prolonged swallow duration
  - Multiple swallow/cough events
  - Irregular sound patterns
  - Weak or noisy signals

Output

- **Risk Level:** Low / Medium / High
- **Explanation:** Based on triggered indicators
- **Guidance:** Safe next steps

What We Analyze

- **Energy** → strength of swallow sound
- **Variability** → consistency vs irregular patterns
- **Sound Events** → multiple peaks (possible coughs)
- **Post-Swallow Activity** → sounds after swallowing
- **Frequency Profile** → unusual sound characteristics
- **Signal Quality** → noise vs clarity
- **ZCR (Zero Crossing Rate)** → turbulence / noisy signals
- **Spectral Features** → frequency characteristics of the sound
- **Bandwidth** → spread of frequencies
- **SNR(signal to noise ratio)** → signal clarity vs background noise

DISCLAIMER

DysphagiaSense is a screening tool only. It does not replace professional medical evaluation. The acoustic analysis is based on general signal processing heuristics and has not been clinically validated. If you have concerns about swallowing difficulties, please consult a healthcare professional or speech-language pathologist.

FUTURE PLANS?